# Three Linux operating modes

A supplied PEM pair takes priority. Only an absent pair selects the configured privileged mechanism.

### Invoke craft as the user

Real UID, NSS name and home
No arguments; no sudo

### Check user.pem and user.key in the NSS home

~/.config/craft/user.pem + user.key
Complete pair takes priority on every installation

BOTH FILES ABSENT

COMPLETE PAIR

Partial, unsafe or invalid:
fail and retain cache.

### Select the configured privileged mechanism

Requires setuid launcher, craft service account and /run/craft

### 1 Privileged enrollment

mechanism=enrollment
Service account: GC + agent + CES
CA issues temporary user certificate
PKINIT authenticates that certificate
CA enforces recipient restrictions

### 2 Privileged credential linking

mechanism=key_trust
Service account: writable DC + keytab
Add temporary msDS-KeyCredentialLink
Key Trust PKINIT; remove the entry
AD enforces attribute-write scope

### 3 Unprivileged Windows PEM

Windows: user enrollment + export
Securely transfer certificate and key
Linux: caller validates and uses pair
PKINIT uses KDC account mapping
Supplied PEM files remain in place

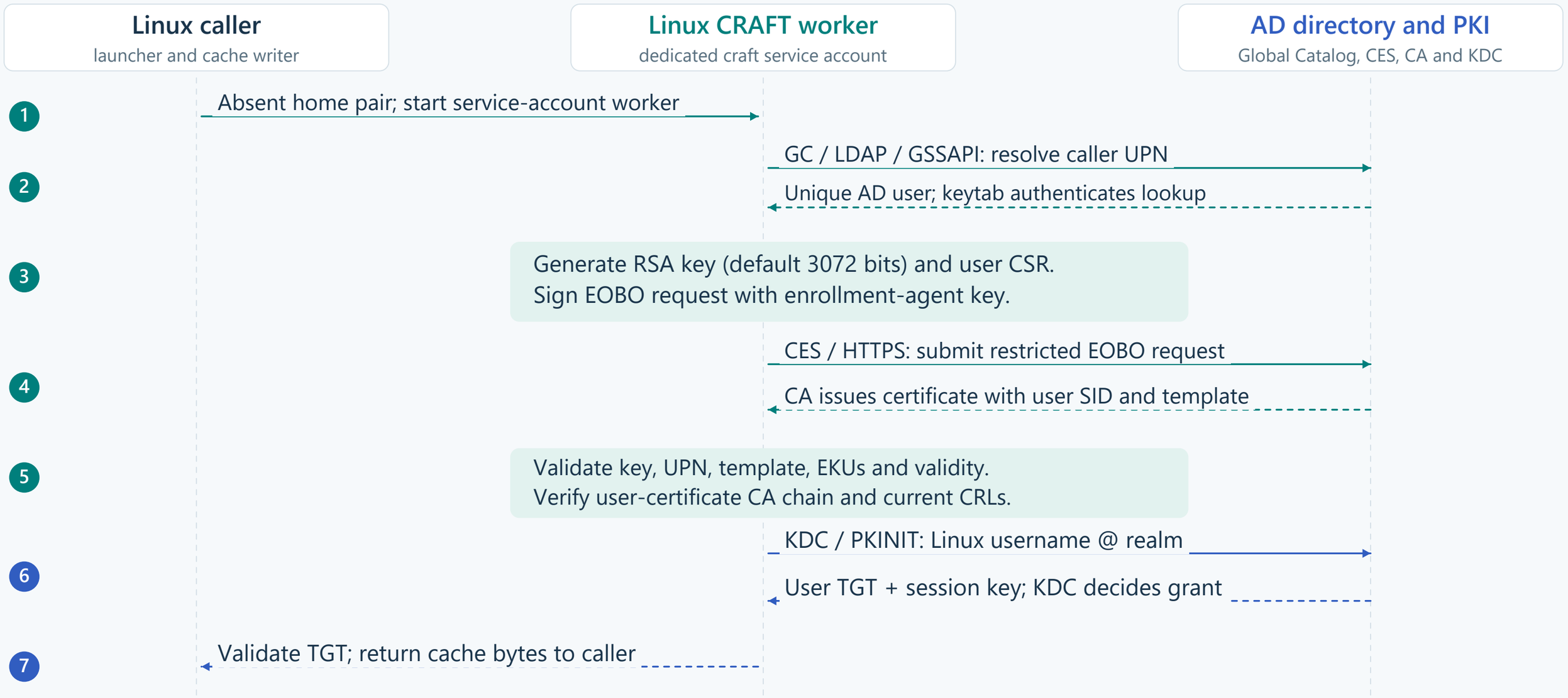
### Validate the user TGT and publish the cache as the caller

Fixed principal: Linux username @ realm; .krb5cc_craft mode 0600; select with KRB5CCNAME

SOURCE source/README.md: Mode selection and process boundaries; mode-specific setup recipes

# Mode 1 Privileged certificate enrollment

Linux orchestrates CA enrollment first, then PKINIT. Select mechanism=enrollment with both home PEM files absent.

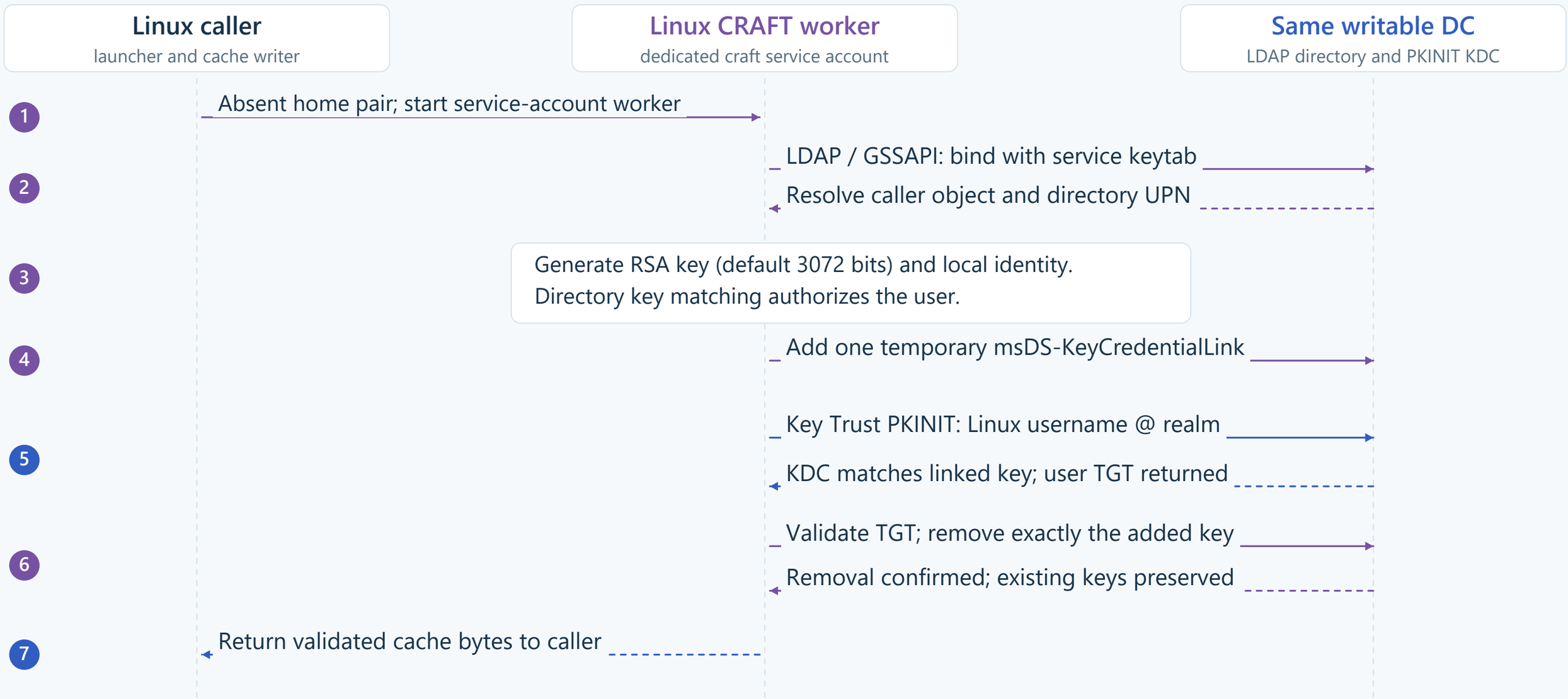



Caller publishes .krb5cc_craft mode 0600. Temporary user key and certificate are released.

SOURCE source/README.md: Mode 1 setup; docs/03-Enrollment-Agent-Restrictions.md

# Mode 2 Privileged credential linking

Linux links a temporary AD public key, performs Key Trust PKINIT, and removes the entry before publishing the TGT.




Caller publishes .krb5cc_craft mode 0600. Cleanup also runs after acquisition failure.
Failed cleanup blocks publication and reports a residual key for administrator removal.

SOURCE source/README.md: Mode 2 setup; docs/04-Key-Credential-Link-Delegation.md

# Mode 3 Unprivileged Windows-exported certificate

Windows enrolls and exports the certificate. Linux uses the transferred PEM pair entirely as the caller.

**Windows domain user**
direct user enrollment and export

**Linux user and CRAFT**
ordinary executables; caller UID

**AD CS and KDC**
certificate issuance and PKINIT

1. Windows enrollment policy: request a user certificate with an exportable software key →
   ← AD CS issues the user certificate; private key stays with the user
2. Request-CRAFT-Certificate.cmd exports user.pem + matching unencrypted user.key
3. Securely transfer both PEM files to Linux →
4. Install at NSS home: ~/.config/craft/user.pem + user.key
   Run craft as the user; check files and certificate.
   Validate CA chain and CRLs; use public /etc/craft policy.
5. KDC / PKINIT: Linux username @ realm →
   ← KDC maps certificate and returns user TGT
6. Validate returned TGT; publish mode-0600 .krb5cc_craft.
   PEM files remain; replace the pair before expiry.

No Linux setuid bit, service account, directory keytab, LDAP lookup, CES or /run/craft.

SOURCE source/README.md: Mode 3 setup; scripts/Request-CRAFT-Certificate.cmd

# Ticket clocks and certificate storage

All three modes request 10 h / 7 d by default. Certificate lifetime and KDC policy constrain the initial grant.

**1 / ENROLLMENT**
Temporary key and certificate.
CA issuance records remain.

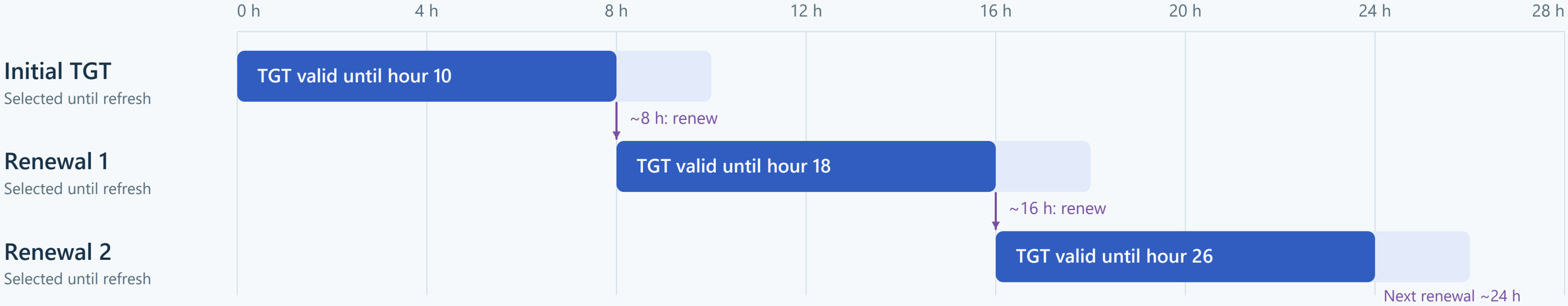
**2 / CREDENTIAL LINKING**
Temporary key and local identity.
Remove directory entry after PKINIT.

**3 / WINDOWS-EXPORTED PEM**
Persistent supplied certificate and key.
Replace the pair before expiry.

01 / HOURS

## Renew before expiry

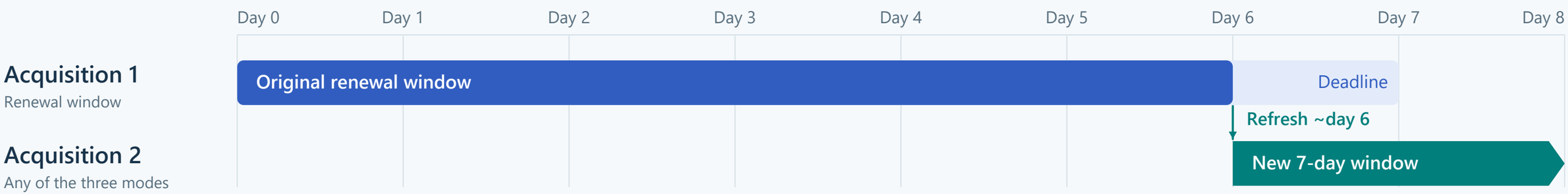
Illustrative full 10 h grants




**Renewal advances expiry; renew-till stays fixed.**

02 / DAYS

## Start a fresh cycle near the renewal deadline




Renewal uses the cached ticket and session key. Certificate expiry or cleanup does not revoke an issued ticket.

SOURCE source/README.md: Configuration reference; Long-running jobs

# Keep a job authenticated, then stop with it.

craft-maintain runs as the caller, coordinates one shared cache per UID, and watches a live process.

START ONCE / WATCH A SHELL THAT STAYS ALIVE FOR THE JOB

```
job_pid=$$
cache=$(/usr/local/bin/craft-maintain --watch-pid "$job_pid") || exit "$?"
export KRB5CCNAME="$cache"
```

Returns when usable credentials and background maintenance are ready.

01 / RENEW

## Use the current ticket

When a renewable TGT has <=20% of its lifetime left (capped at 2 h), and its expiry can extend.

A renewal error retries; no immediate fresh authentication.

**Cached TGT** + session key → **KDC** renew request → **Validated TGT** new expiry

Renew-till never moves later.
Retain usable service tickets.

02 / FRESH AUTHENTICATION

## Obtain fresh credentials

Missing or expired TGT; renewal deadline is near; or a ticket that cannot extend is nearing expiry.

Rollover margin: 1/7 of the renewal window, capped at 1 day.

**Run craft** select source → **Selected mode** 1, 2 or 3 → **KDC / PKINIT** new user TGT

Home PEM, enrollment or credential linking.
New renewal window; replace cache.

A due renewal takes priority during rollover, keeping usable credentials alive while fresh authentication is delayed.

### Successful refresh

Publish atomically to .krb5cc_craft (caller-owned, mode 0600).

### Refresh failure

Keep the existing cache and retry; the job continues.

RETRY

Renewal: ~30 s to 5 min + jitter.
Fresh acquisition: 1 min to 1 h cooldown.

STOP

Watched PID exits, duration ends or signal.
Maintainer stops; the shared cache remains.

APPLICATION

Reload credentials when authenticating again.
Access follows the user's service permissions.

SOURCE source/README.md: Long-running jobs; source/SECURITY.md: Immediate disable